

ABPN: Apex and Boundary Perception Network for Micro- and Macro-Expression Spotting

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ABSTRACT

Recently, Micro expression (ME) has achieved remarkable progress in a wide range of applications, since it's an involuntary facial expression that reflects personal psychological state truly. In the procedure of ME analysis, spotting ME is an essential step, and is non trivial to be detected from a long interval video because of the short duration and low intensity issues. To alleviate this problem, in this paper, we propose a novel Micro- and Macro-Expression (MaE) Spotting framework based on Apex and Boundary Perception Network (ABPN), which mainly consists of three parts, i.e., video encoding module (VEM), probability evaluation module (PEM), and expression proposal generation module (EPGM). Firstly, we adopt Main Directional Mean Optical Flow (MDMO) algorithm and calculate optical flow differences to extract facial motion features in VEM, which can alleviate the impact of head movement and other areas of the face on ME spotting. Then, we extract temporal features with one-dimension convolutional layers and introduce PEM to infer the auxiliary probability that each frame belongs to an apex or boundary frame. With these frame-level auxiliary probabilities, the EPGM further combines the frames from different categories to generate expression proposals for the accurate localization. Besides, we conduct comprehensive experiments on MEGC2022 spotting task, and demonstrate that our proposed method achieves significant improvement with the comparison of state-of-the-art baselines on CAS(ME)² and SAMM-LV datasets. The implemented code is also publicly available at https://github.com/wenhaocold/USTC_ME_Spotting.

CCS CONCEPTS

• Computing methodologies → Computer vision tasks.

*Wenhao Leng and Sirui Zhao contributed equally to this research. Tong Xu and Enhong Chen are the corresponding authors.

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MM '22, October 10–14, 2022, Lisboa, Portugal

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ACM ISBN 978-1-4503-9203-7/22/10...\$15.00

<https://doi.org/10.1145/3503161.3551599>

KEYWORDS

micro- and macro- expression, optical flow, convolutional neural networks

ACM Reference Format:

Wenhao Leng, Sirui Zhao, Yiming Zhang, Shifeng Liu, Xinglong Mao, Hao Wang, Tong Xu, and Enhong Chen. 2022. ABPN: Apex and Boundary Perception Network for Micro- and Macro-Expression Spotting. In *Proceedings of the 30th ACM International Conference on Multimedia (MM '22)*, October 10–14, 2022, Lisboa, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3503161.3551599>

1 INTRODUCTION

In recent years, Micro expression has made remarkable progress and has been gradually applied to extensive applications, such as criminal investigation, national security, clinical psychology, and other fields [4, 14, 15, 20], since it's spontaneous and uncontrolled that provides effective clues to reveal people's true psychological state even if they attempt to conceal it [5, 6, 10].

However, it is difficult for human to spot and recognize MEs due to their short duration and low intensity. Although Ekman developed the ME Training Tool (METT) to train human's ability to analyze MEs [3], its performance is still far below a desirable level[7]. Therefore, it is essential to develop automatic ME analysis systems.

Generally, automatic ME analysis consists of two essential steps: ME spotting [19, 22, 25, 26] and ME recognition [9, 21, 28, 29]. ME spotting is to locate the start and end of ME clips from a long video; ME recognition is to classify the emotion of ME clips. At present, the work on ME spotting is relatively rare compared with ME recognition. However, as the first step in ME analysis, ME spotting is crucial for the subsequent ME recognition task.

In this paper, we focus on the ME spotting task and propose a ME spotting network that perceives apex and boundary frames, named Apex and Boundary Perception Network (ABPN). The network takes the optical flow based features as input. Because of short duration and low intensity, ME is vulnerable to unrelated interferences. For example, eye blinks, head movements and deformations of the face region unrelated to ME will interfere in the spotting result. Hence, in the video encoding model (VEM), we use the MDMO algorithm to evaluate optical flows of ROIs related to ME to alleviate the interference and use optical flow differences[27] to eliminate head movement. Then, the ABPN, which refers to the method proposed in BSN[11], takes the video features as input to generate expression proposals. Nevertheless, unlike BSN, our method evaluates the probabilities for each frame in a video, and the sequential step can use

the frame-level probabilities to form expression proposals, which have a high IOU with ground truth intervals. Besides, our method focuses on the apex frame, which is unique to the expression clip and has the highest intensity compared with other frames, and uses start and end frames as auxiliary. The final result shows that our method achieves competitive results on MEGC2022-Testset.

In summary, the main contributions of our work are three-folds: 1) We use the manually extracted features to get expression information and alleviate unrelated interference; 2) We improve the method of BSN so that it can be applied to the ME spotting task properly. 3) We demonstrate the efficacy of the proposed method in terms of the F1-score on MEGC2022-Testset.

2 RELATED WORK

Current ME spotting methods can be roughly divided into two categories, traditional methods and deep-learning based methods. For the traditional methods, manually extracted features are used for spotting MEs. For example, Moilanen et al.[13] adopted the Local Binary Pattern (LBP) algorithm to extract facial features and then used feature-difference to locate apex frames. Zhang et al. [27] used the optical flow algorithm to extract the motion information between frames and then constructed a sequence of optical flows to get the spatio-temporal features and then extracted SP-patterns from features to obtain expression intervals. For deep-learning methods, 2D-CNNs are used to extract spatial features, and RNNs or 1D-CNNs are used to extract temporal features. For example, Wang et al.[19] adopted 2D-CNNs and 1D-CNNs to extract spatio-temporal features of videos and then used five 1D-CNNs with different receptive fields to generate expression proposals with different scales, and finally used regression network to fine-tune the localization of expression proposals. Yap et al.[24] imposed 3D-CNNs as the basic structure of the network to extract spatio-temporal features and then used long-short skip to take advantage of the major difference between ME and MaE durations.

3 METHODOLOGY

To accurately detect the MEs and MaEs from long videos, we propose a novel ME and MaE Spotting framework based on Apex and Boundary Perception Network (ABPN), which consists of three modules:

- **Video Encoding Module (VEM).** This module mainly focuses on extracting spatial motion features from the long video based on optical flow motion estimation.
- **Probability Evaluation Module (PEM).** The purpose of the probability evaluation module is to evaluate the proper category that a frame should be.
- **Expression Proposal Generation Module (EPGM).** The expression proposal generation module aims to generate proposals for ME and MaE according to the probabilities of each frame's category.

3.1 Video Encoding Module

In this module, we first adopt retinaface[1] and SAN[2] to extract the coordinates and landmarks of the face, respectively, in preparation for the subsequent calculation of the optical flows of ROIs

related to the action unit. We select 12 regions based on landmarks that might be activated when micro-expression occurs[27].

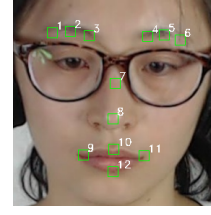


Figure 1: Regions of Action Unit related to ME

Secondly, we use the TV-L1 optical flow method[17] to calculate the optical flows of ROIs, and adopt the MDMO[12] method to obtain the optical flows in the main direction. Then, we calculate the sum of the main directional optical flows of each ROI and the unit vector of that sum, and use the ratio of the two as the feature of the region.

$$\bar{\mathbf{u}}^{roi} = \frac{\sum \mathbf{u}_i^{roi}}{\|\sum \mathbf{u}_i^{roi}\|_2}, \quad \mathbf{u}_i^{roi} \in R_{main}$$

where R_{main} is a set of optical flows in the main direction.

In order to alleviate the impact caused by head movement on ME spotting, we calculate the global optical flow to represent the head movement. As the motion of the nose region can be regarded as a rigid body motion compared with other action units, we can utilize the optical flow of the nose region to represent the global optical flow. As before, the global optical flow feature $\mathbf{u}_g$ is obtained by calculating the MDMO of the nose region to represent the head movement. Then the optical flow feature of each ROI area is subtracted from the global optical flow feature, which can effectively eliminate the movement of the head. So we can get the feature of each ROI area,

$$\mathbf{f}^{roi} = \bar{\mathbf{u}}^{roi} - \mathbf{u}_g$$

and a feature matrix $\mathbf{X}$ with dimension $(N-1) \times 12$ is obtained to represent the features of a video with length N .

$$\mathbf{X} = \begin{bmatrix} F_1 \\ F_2 \\ \vdots \\ F_{N-1} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_1^0 & \mathbf{f}_1^1 & \cdots & \mathbf{f}_1^{11} \\ \mathbf{f}_2^0 & \mathbf{f}_2^1 & \cdots & \mathbf{f}_2^{11} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{f}_{N-1}^0 & \mathbf{f}_{N-1}^1 & \cdots & \mathbf{f}_{N-1}^{11} \end{bmatrix}$$

3.2 Probability Evaluation Module

As stated above, this module evaluates the probabilities of each frame, indicating whether a frame is inside or outside of an expression, and whether a frame is a special frame, including the apex and boundary frame, or a normal frame. These correspond to three classification tasks: a binary classification task of estimating whether a frame is an apex frame; a multi-classification task of estimating whether a frame is a start frame, end frame, or normal frame; a binary classification task of estimating whether a frame belongs to an expression clip. We apply two groups of these tasks to distinguish ME and MaE.

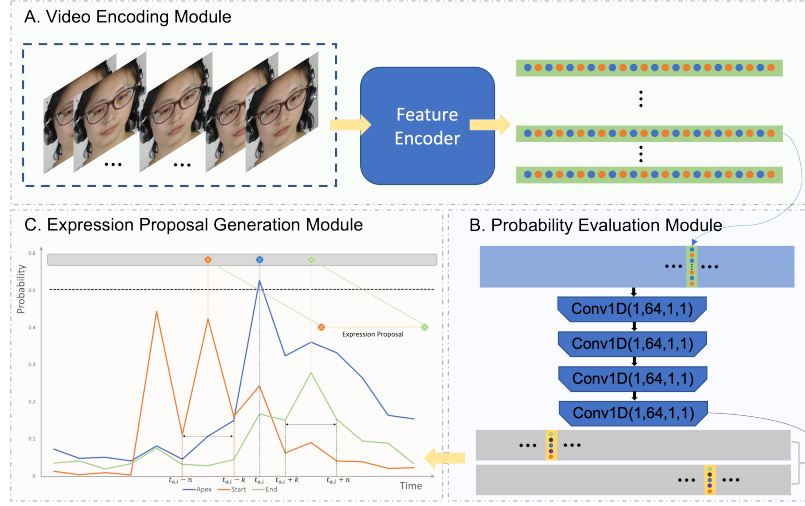


Figure 2: The architecture of ABPN. (A) MDMO algorithm and optical flow difference are used to encode video features. (B) probability evaluation module utilizes the input features to evaluate the probabilities of each frame. (C) expression proposal generation module utilizes the probabilities of each frame to generate expression proposals

As shown in Figure 2, the probability evaluation module consists of four one-dimension convolutional layers, which takes the aforementioned features in section 3.1 as input. To make the model learn the temporal information of expressions, we use the one-dimension convolutional layer as the main component of the network. The convolutional layer can be denoted as $Conv1D(k, c_{out}, p, d)$, where k is the kernel size, c_{out} is the feature dimension of the layer output, p is the padding size, and d is the dilation size. We also apply a one-dimension batch normalization layer and Relu activation function after each convolutional layer except for the last, as it dedicates to classification tasks. For the binary classification task, we apply sigmoid as the activation function; for the three-class classification task, we apply softmax as the activation function. We set stride as 1 to keep the temporal length of output the same as input, while, in order to increase the receptive field of the model without increasing the complexity of the model's parameters, we use dilation convolution in the last two layers.

Formally, given a video feature X with length l , PEM generates ten probabilities for each frame, divided into two groups for ME and MaE spotting respectively, so that we can get two groups of probability sequences for a video. Each group uses the following symbols to represent probability sequences: $P_{start} = \{p_t^{start}\}_{t=1}^l$, $P_{end} = \{p_t^{end}\}_{t=1}^l$, $P_{apex} = \{p_t^{apex}\}_{t=1}^l$, $P_{normal} = \{p_t^{normal}\}_{t=1}^l$, $P_{exp} = \{p_t^{exp}\}_{t=1}^l$, where p_t^{start} , p_t^{end} , p_t^{apex} are probabilities of the start, end, and apex frame respectively and where p_t^{normal} , p_t^{exp} are probabilities of the normal frame and expression frame.

3.3 Expression Proposal Generation Module

The purpose of the expression proposal generation module is to generate proposals according to the probabilities of each frame's category. The apex frame has the highest intensity in a ME clip and can be spotted easily compared with other frames. Therefore, first,

we compare the apex probability of each frame obtained by PEM with the threshold condition, $p_t^{apex} > threshold_{apex}$, to get the candidate set of apex frames $C_a = \{t_{a,i}\}_{i=1}^{N_a}$, where $t_{a,i}$ represents the location of the candidate apex frames in a video, and N_a represents the number of candidate apex frames. The probability sequence of start frames and end frames around a candidate apex frame $t_{a,i}$ are represented as $P_{t_{a,i}}^{start} = \{p_{t_{a,i}-n}^{start}, p_{t_{a,i}-n+1}^{start}, \dots, p_{t_{a,i}-k}^{start}\}$ and $P_{t_{a,i}}^{end} = \{p_{t_{a,i}+k}^{end}, \dots, p_{t_{a,i}+n-1}^{end}, p_{t_{a,i}+n}^{end}\}$ respectively, where n and k are half of the average length and minimum length of ME in the dataset. Then, we select the start and end frame $t_{s,i}$, $t_{e,i}$ with the highest probability from $P_{t_{a,i}}^{start}$ and $P_{t_{a,i}}^{end}$ respectively to form the candidate set of start frames $C_s = \{t_{s,i}\}_{i=1}^{N_a}$ and end frames $C_e = \{t_{e,i}\}_{i=1}^{N_a}$. Lastly, we filter the start and end candidate set with the condition, $p_{t_{s,i}}^{start} > p_{t_{s,i}}^{normal}$ and $p_{t_{e,i}}^{end} > p_{t_{e,i}}^{normal}$, to generate expression proposals $EP = \{[t_{s,i}, t_{e,i}]\}_i^{N_{ep}}$, where N_{ep} is the number of expression proposals.

3.4 Optimization

We get the output $\hat{Y} \in R^{256 \times 10}$ from PEM, which takes features of a clip $C \in R^{256 \times 24}$ as input. Probabilities generated for each frame in $\hat{Y}$ need two groups of loss functions to optimize it for ME and MaE, respectively. And each of the groups has the same number and type of loss function. The output for binary classification task in $\hat{Y}$ is denoted as $\hat{y}_b$ and the corresponding true label is denoted as y_b . We use the following loss to optimize the task.

$$L_b = -[\alpha(1-y)y \log \hat{y} + (1-\alpha)y^y(1-y) \log(1-\hat{y})] + \frac{\lambda}{2} \sum_w w^2$$

The output for three-class classification task in $\hat{Y}$ is denoted as $\hat{y}_m = (\hat{y}_1, \hat{y}_2, \hat{y}_3)^T$ and the corresponding true label is denoted as

Table 1: F1-score comparison with the state-of-the-art models on CAS(ME)² and SAMM-LV

Method	CAS(ME) ²			SAMM		
	MaE	ME	ALL	MaE	ME	ALL
Baseline[24]	0.2145	0.0714	0.1675	0.1595	0.0466	0.1084
He et al.[8]	0.1196	0.0082	0.0376	0.0629	0.0364	0.0445
Zhang et al.[27]	0.2131	0.0547	0.1403	0.0725	0.1331	0.0999
Ours(no-EPGM)	0.2717	0.1429	0.2541	0.2330	0.1498	0.2095
Ours	0.3357	0.1590	0.3117	0.3349	0.1689	0.2908

$\mathbf{y}_m = (y_1, y_2, y_3)^T$. We use the following loss to optimize the task.

$$L_m = - \sum_{i=1}^3 \alpha_i (1 - y_i)^y y_i \log \hat{y}_i + \frac{\lambda}{2} \sum_w w^2$$

The loss function for the expression spotting consists of the apex loss, start-end-normal classification loss and expression loss:

$$L_{Ex} = \lambda_1 L_{b-apex} + \lambda_2 L_{m-start_end} + \lambda_3 L_{b-expression}$$

And the overall loss function for ME and MaE spotting is following:

$$L_{PEM} = L_{Ex-Micro} + \lambda L_{Ex-Macro}$$

4 EXPERIMENTS

4.1 Datasets

We evaluate our method on the test set of MEGC2022, which contains ten long videos, including five long videos from SAMM dataset, and five clips cropped from different videos in CAS(ME)³[10]. The frame rate for SAMM Challenge dataset is 200fps and the frame rate for CAS(ME)³ is 30 fps. To extend the dataset for training PEM, We construct a new dataset named Cross-ME, consisting of CAS(ME)²[16], SAMM[23] and SMIC-E-Long[18]. CAS(ME)² has 98 long videos with 30 fps and contains 57 ME samples and 300 MaE samples. SAMM has 224 long videos with 200 fps and contains 159 ME samples and 343 MaE samples. SMIC-E-Long has 162 long videos with 100 fps and contains 166 ME samples. Besides, we divide the Cross-ME dataset into training and validation sets by 8:2 and use the validation set to tune the model's hyperparameters.

4.2 Implementation Detail

To ensure all samples in Cross-ME datasets have the same frame rate, we downsample the SAMM-Long and SMIC-E-Long datasets from their original frame rate to 25fps. Because the frame rate of CAS(ME)² is 30fps, which is close to 25fps, we keep it untouched. For the parallelization during training PEM, we divide each video into clips with the same length of 256 while excluding clips that do not contain ME or MaE frames to avoid the normal frames overwhelming expression frames.

4.3 Evaluation Metric

We use the performance metrics proposed in [24] to evaluate the final results. For the proposal generated by our model, it is considered as true positive if the following condition is satisfied:

$$\frac{W_{\text{proposal}} \cap W_{\text{groundTruth}}}{W_{\text{proposal}} \cup W_{\text{groundTruth}}} \geq T_{\text{eval}}$$

where T_{eval} is the threshold and is set to 0.5. For a ground truth interval, it corresponds to at most one true positive proposal, which is named hit-proposal. Then the recall, precision and f1-score are calculated as follows:

$$\begin{aligned} \text{Recall} &= \frac{a}{m}, & \text{Precision} &= \frac{a}{n} \\ \text{F1-score} &= \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} = \frac{2a}{n + m} \end{aligned}$$

where a , m and n are the numbers of hit-proposals, ground truth intervals and all predicted proposals, respectively.

Table 2: F1-score on unseen test set of MEGC2022

Method	CAS(ME) ³	SAMM Challenge	Overall
Baseline[24]	0.1739	0.1176	0.1351
Ours(no-EPGM)	0.2358	0.1429	0.1970
Ours	0.3529	0.2264	0.2975

4.4 Experimental Results

To fully verify the effectiveness of our proposed ABPN model, we conducted experiments in accordance with MEGC2022 and MEGC2021 rules respectively, and compared the spotting results with other SOTA methods. The corresponding results are presented in Table2 and Table1. The no-EPGM in the table means only using the candidate apex frames and fixed length to generate proposals. From the tables, it's obvious that our ABPN can achieve surprising results, which fully proves the effectiveness of our method.

5 CONCLUSION

In this paper, We proposed a novel ME and MaE spotting method based on Apex and Boundary Perception Network (ABPN). To verify the proposed ABPN method, we conducted comprehensive experiments on CAS(ME)², SAMM-LV and SMIC-E-long datasets, in which the experimental results show that our method achieves competitive results on MEGC2022 spotting task.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Found of China (Grant No. 61727809, 62076209).

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